

112590024 邱紹澹 資工三

112590004 邱昱綸 資工三

CH4 Project 1

執行後可以正確判斷。

```
kelen@lol:~/Documents/GitHub/OSHW/HW2/OSHW_2_teambased_112590024_112590004/CH4_1$ go build sudokuParams.go
kelen@lol:~/Documents/GitHub/OSHW/HW2/OSHW_2_teambased_112590024_112590004/CH4_1$ ./sudokuParams
Valid Grid:
[5 3 4 6 7 8 9 1 2]
[6 7 2 1 9 5 3 4 8]
[1 9 8 3 4 2 5 6 7]
[8 5 9 7 6 1 4 2 3]
[4 2 6 8 5 3 7 9 1]
[7 1 3 9 2 4 8 5 6]
[9 6 1 5 3 7 2 8 4]
[2 8 7 4 1 9 6 3 5]
Valid Grid: Sudoku solution is VALID
Invalid Grid:
[5 3 4 6 7 8 9 1 5]
[6 7 2 1 9 5 3 4 8]
[1 9 8 3 4 2 5 6 7]
[8 5 9 7 6 1 4 2 3]
[4 2 6 8 5 3 7 9 1]
[7 1 3 9 2 4 8 5 6]
[9 6 1 5 3 7 2 8 4]
[2 8 7 4 1 9 6 3 5]
[3 4 5 2 8 6 1 7 9]
Invalid Grid: Sudoku solution is INVALID
```

ch4_p1_print.png

CH4 Project 2

兩個thread啟動，左半先完成後，換右半完成

```
kelen@host:~/文件/GitHub/OSHW/HW2/OSHW_2_teambased_112590024_112590004/CH4_2$ ./a.out
請輸入整數個數: 10 multi_sort.c
請輸入 10 個整數 (以空格分隔) : 1 5 8 9 20 8 7 5 4 3
排序前: [ 1 5 8 9 20 8 7 5 4 3 ]
切割方式:
左半部 index 0 ~ 4 (5個元素)
右半部 index 5 ~ 9 (5個元素)
啟動執行緒 ...
[執行緒] 開始排序 index 0 ~ 4
[執行緒] 開始排序 index 5 ~ 9
[執行緒] 完成排序 index 5 ~ 9
[執行緒] 完成排序 index 0 ~ 4
兩個執行緒皆已完成
左半部排序後: [ 1 3 4 5 7 8 ]
右半部排序後: [ 3 4 5 7 8 ]
主執行緒執行最終合併...
排序後: [ 1 3 4 5 5 7 8 8 9 20 ]
程式結束。
```

CH5 Project 1

```
CH5$ ./a.out schedule.txt 1

--- FCFS Scheduling ---
Running task = [T1] [Priority: 4] for 20 units.
Running task = [T2] [Priority: 2] for 25 units.
Running task = [T3] [Priority: 3] for 25 units.
Running task = [T4] [Priority: 3] for 15 units.
Running task = [T5] [Priority: 10] for 10 units.
kelen@host:~/文件/GitHub/OSHW/HW2/OSHW_2_teambase
CH5$ ./a.out schedule.txt 2

--- SJF Scheduling ---
Running task = [T5] [Priority: 10] for 10 units.
Running task = [T4] [Priority: 3] for 15 units.
Running task = [T1] [Priority: 4] for 20 units.
Running task = [T2] [Priority: 2] for 25 units.
Running task = [T3] [Priority: 3] for 25 units.
kelen@host:~/文件/GitHub/OSHW/HW2/OSHW_2_teambase
CH5$ ./a.out schedule.txt 3

--- Priority Scheduling ---
Running task = [T5] [Priority: 10] for 10 units.
Running task = [T1] [Priority: 4] for 20 units.
Running task = [T4] [Priority: 3] for 15 units.
Running task = [T3] [Priority: 3] for 25 units.
Running task = [T2] [Priority: 2] for 25 units.
```

CH5_p1_1.png

```
kelen@host:~/文件/GitHub/OSHW/HW2/OSHW_2_teambased_112
CH5$ ./a.out schedule.txt 4

--- Round-Robin Scheduling (Quantum = 10) ---
Running task = [T1] [Priority: 4] for 10 units.
Running task = [T2] [Priority: 2] for 10 units.
Running task = [T3] [Priority: 3] for 10 units.
Running task = [T4] [Priority: 3] for 10 units.
Running task = [T5] [Priority: 10] for 10 units.
Running task = [T1] [Priority: 4] for 10 units.
Running task = [T2] [Priority: 2] for 10 units.
Running task = [T3] [Priority: 3] for 10 units.
Running task = [T4] [Priority: 3] for 5 units.
Running task = [T2] [Priority: 2] for 5 units.
Running task = [T3] [Priority: 3] for 5 units.
kelen@host:~/文件/GitHub/OSHW/HW2/OSHW_2_teambased_112
CH5$ ./a.out schedule.txt 5

--- Priority with RR Scheduling (Quantum = 10) ---
Running task = [T5] [Priority: 10] for 10 units.
Running task = [T1] [Priority: 4] for 10 units.
Running task = [T1] [Priority: 4] for 10 units.
Running task = [T4] [Priority: 3] for 10 units.
Running task = [T3] [Priority: 3] for 10 units.
Running task = [T4] [Priority: 3] for 5 units.
Running task = [T3] [Priority: 3] for 10 units.
Running task = [T3] [Priority: 3] for 5 units.
Running task = [T2] [Priority: 2] for 10 units.
Running task = [T2] [Priority: 2] for 10 units.
Running task = [T2] [Priority: 2] for 5 units.
```

CH5_p1_2.png